CIP HS26
Feasibility Study
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Premiums vs. Paychecks: Health Insurance Affordability Across Swiss Cantons, 2012–2022

1. Introduction, scope and purpose

Mandatory health insurance premiums in Switzerland have increased substantially over the past decade, reflecting continued growth in healthcare expenditure (Federal Office of Public Health FOPH, 2026b). Because the premium is charged per person and is not scaled to income, the same amount weighs differently on different households. This project compares premiums with income across the 26 Swiss cantons between 2012 and 2022 to measure that weight and to show how it has developed.

The main objective is to compare annual health insurance premiums with taxable income at cantonal level and to examine how this relationship has changed over time. The project also looks at the differences between the Swiss cantons and at whether those differences have grown or narrowed over the study period, which runs from 2012 to 2022 because the cantonal tax statistics currently end with tax year 2022.

The analysis combines publicly available premium, tax and other statistical data from Swiss federal sources. The data are collected with web scraping and an API, then cleaned and combined in Python, and finally analyzed and visualized. Selenium, BeautifulSoup and Requests are used for the acquisition, Pandas and NumPy for the processing, and Matplotlib and Plotly for the visualization.

2. Research questions

RQ1: Change in affordability over time: How has the affordability of mandatory health insurance changed in Switzerland between 2012 and 2022?

RQ1.1: Did health insurance premiums increase faster than income over this period?

RQ2: Regional differences: How does the affordability of mandatory health insurance differ between Swiss cantons, and have these regional differences changed between 2012 and 2022?

RQ3: Effect of insurance choices: How much do health insurance premiums differ by deductible and insurance model across the cantons?

3. Data sources and acquisition

All sources are public federal sources. Details and all links are in the appendix.

Method	Source	Data	Python tools
API	BFS STAT-TAB PxWeb API (endpoint)	Median gross wage by large region 2012–2024	requests (JSON POST queries), pandas
Scraping (dynamic)	Priminfo premium calculator (FOPH)	Current premiums per canton/region, age group, deductible, model, insurer	Selenium, BeautifulSoup
Scraping (static)	Priminfo archive	Yearly premium and subsidy PDFs per canton, 2012–2025	BeautifulSoup (link list), requests, pdfplumber
Scraping (static)	ESTV direct federal tax statistics	Taxable income and taxpayers per canton, 1983–2022 (yearly XLSX)	BeautifulSoup (file links), requests, pandas.read_excel
Backup	opendata.swiss – BAG premiums	Full premium tables (CSV) per insurer and region	Manual, requests download, pandas.read_csv

4. Planned cleaning, transformation and analysis

Income data preparation: The income data are available at canton level and give the mean and the median taxable income per canton, split into married, unmarried and total taxpayers. The median of the total row is used, because income is skewed to the right. The yearly files are stacked into one table before being combined with the premium data.

Premium data preparation: Premium data will be collected with Selenium for all 26 cantons and multiple years. The extracted data will be filtered according to age group, deductible and insurance model depending on the analysis.

Data validation: Premium and income values will be checked for missing, zero or negative values and for obvious inconsistencies. Unusual values will be investigated before deciding whether they should be excluded. No fixed outlier threshold will be defined in advance.

Derived variables: New variables will be calculated from the collected data to compare health insurance premiums with income. These may include a premium-to-income ratio, changes in premiums and income over time, and differences between deductible or insurance model options.

Analysis: Premium and income development will be compared across cantons and over time. Depending on the final structure and quality of the income data, the relationship between income and premiums may additionally be investigated using regression analysis.

Visualization: Results will be presented using line plots to show changes over time, bar plots for comparisons between cantons or insurance options, and a choropleth map of Swiss cantons to visualize geographical differences in affordability. A heatmap may additionally be used to compare cantons across multiple years.

5. Risks and mitigation

Risk	Mitigation / backup plan
Websites may block automated access or change its website structure, causing the Selenium scraper to fail.	Add a delay between requests to reduce server load. If scraping fails, use manually downloaded CSV data from opendata.swiss as a backup source
Premium data provided in PDF tables may be difficult to extract reliably.	Extract the tables using pdfplumber and perform year-by-year plausibility checks. Validate the extracted values against the annual mean premiums published by BAG
Data is available in different formats and structures	Clean and standardize the datasets before using them for analysis
Data contain missing values	Identify and flag missing values. Interpolation will only be used where appropriate and will be documented
Premium data contain many combinations of age, deductible and insurance model	Filter required data according to the research question we focus on
API or website structure changes	Save raw downloads in data/raw/ with the retrieval date; scripts can be re-run

6. References

Federal Office of Public Health (FOPH). (2026a). *Health insurance: premium regions*. Retrieved October 3, 2026, from https://www.bag.admin.ch/de/krankenversicherung-praemienregionen

Federal Office of Public Health (FOPH). (2026b). *Prämienrechner 2027 – Priminfo*. Priminfo. Retrieved October 3, 2026, from https://www.priminfo.admin.ch/de/praemien

Federal Office of Public Health (FOPH). (2026c). *Statistics on compulsory health insurance*. Federal Office of Public Health FOPH. Retrieved October 3, 2026, from https://www.bag.admin.ch/en/statistics-on-compulsory-health-insurance

Federal Office of Public Health (FOPH). (2026d). *Health insurance premiums - opendata.swiss* [Dataset]. Retrieved October 3, 2026, from https://opendata.swiss/en/dataset/health-insurance-premiums

Federal Statistical Office (BFS). (2026). *STAT-TAB - interaktive Tabellen (BFS)* [Dataset]. Federal Statistical Office. https://www.pxweb.bfs.admin.ch/api/v1/de/

Federal Tax Administration (FTA). (2026). Statistics on Direct Federal tax, natural persons [Dataset]. In *Federal Tax Administration FTA*. Retrieved October 3, 2026, from https://www.estv.admin.ch/de/statistiken-zur-direkten-bundessteuer

7. Appendix - Source links and access check

Type	Source	Link	Status
API	BFS STAT-TAB PxWeb API (endpoint)	https://www.pxweb.bfs.admin.ch/api/v1/de/	No robots.txt on this host, so nothing is disallowed. Endpoint returns JSON. CPI and population tables still to be selected.
Scraping	Priminfo premium calculator	https://www.priminfo.admin.ch/de/praemien	No robots.txt on this host, so nothing is disallowed. Page loads and the form is JavaScript, reached with Selenium.
Scraping	Priminfo archive (yearly PDFs)	https://www.priminfo.admin.ch/de/downloads/archiv	No robots.txt on this host, so nothing is disallowed. Link list parsed, premiums 1996 to 2026 and subsidies 2015 to 2025
Scraping	Example: premiums 2025 PDF	https://www.priminfo.admin.ch/downloads/praemien_2025.pdf	No robots.txt on this host, so nothing is disallowed. Opens and holds premium tables for all 26 cantons by deductible, model and premium region.
Scraping	Example: subsidy split 2025 PDF	https://www.priminfo.admin.ch/downloads/praemienverbilligung_aufteilung_2025_de.pdf	No robots.txt on this host, so nothing is disallowed. Opens and holds one row per canton with population and the federal contribution
Scraping	ESTV direct federal tax statistics (overview)	https://www.estv.admin.ch/de/statistiken-zur-direkten-bundessteuer	robots.txt allows access. Page lists the yearly XLSX files for 1983 to 2022.
Scraping	ESTV natural persons by canton 1983–today	https://www.estv.admin.ch/de/statistiken-natuerliche-personen-kantone-1983-bis-heute	robots.txt allows access. Page lists one yearly XLSX per canton.
Backup	opendata.swiss – Health insurance premiums (BAG)	https://opendata.swiss/en/dataset/health-insurance-premiums	Open use, source must be credited. downloaded through the browser.